1. Bevel Angle Range 37.5 with tolerance +- 2.5?

    a. 36-42
    b. 35-40
    c. 33-38
    d. 35.5-39.5

2. Find out Range of 25% of 20.
    a. 3
    b. 2
    c. 5
    d. 8

3. A welder continuously held for 10 minutes with 60 duty cycle to a complete weld by using 100% duty cycle, how much Minutes it will take?

    a. 10 Minutes
    b. 6 Minutes
    c. 8 Minutes
    d. 15 Minutes

4. 40 circumferential of a pipe tolerance 20%, what the total tolerance rate?

5. Qualification Range 0.5 to 1.1T. The value of the T is 0.76. What is the range?

    a. 0.38 to 0.836
    b. 0.46 to 0.789
    c. 0.34 to 0.658
    d. 0.58 to 0.936

6. For 20mm given +-10% tolerance. What is the tolerance range?

    a. 16 to 23
    b. 19 to 22
    c. 18 to 22
    d. 18 to 21

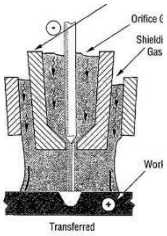
7. Root Opening is 1/8” allowable tolerance range is +1/16” and -1/32”, what is the range?

    3/32 to 3/16

8. For 20” Pipe 40% is welding complete. What is the length for completed welding?
    a. 26
    b. 27
    c. 25
    d. 28

9. The most Extreme used NDT Method?
   a. UT
   b. VT
   c. RT
   d. MT

10. Which is the correct as shown figure?



   a. GTAW
   b. SMAW
   c. FCAW
   d. PAW

11. Proper means correcting record?
   a. Complete white out incorrect entry, re-enter, initial and date
   b. Complete black out incorrect entry, re-enter, initial and date
   c. Line out incorrect entry with a single line, correct, date and initial
   d. All of the above

12. Which one is correct as show figure?
   a. Baking
   b. Back welding
   c. Melt trough
   d. Grind Flash

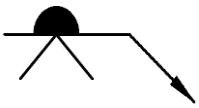
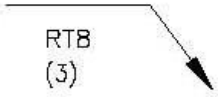


Fig. 10

13. The Symbol RT (3) means?
   a. RT for 8” at 3 random location
   b. 8 RT every 3”
   c. 3” long RT for every 8”
   d. 3RT for eith pitch of 8”



14-Carbon Equivalent can found which of the following? a. Martensitic
   b. Austenite
   c. Perlite
   d. Ferrite

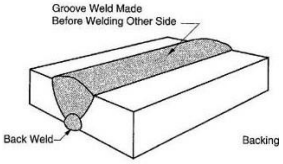
15-Special Specification Condition? a. PQR
b. WQT
c. WPS
d. Drawing

16. FCAW-S, so the S stand for?

a. Gas shielded
b. Self-shielded

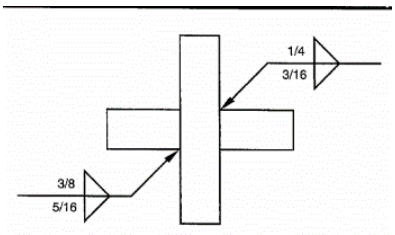
17. Which one is correct as shown figure?

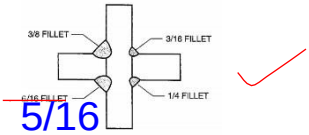
a. Backing Weld
b. Back Weld
c. Melt through
d. Bead weld



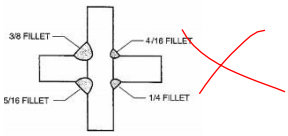
18. Definition of Weld Toe?

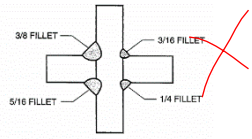
19. Which one is correct as per the Figure?





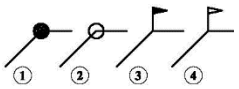
5/16





20. Which of the symbols as listed in the sketch, is used it indicate a field weld?

a. 1
b. 2
c. 3
d. 4



21. A weld with 1” thickness, 2nd layer completed with 5/16” and Final layer completed with ½” with no additional reinforcement required, what is the thickness for root pass?

    a. 6/16”
    b. 4/8”
    c. 3/16”
    d. 3/8”

22. In Radiography testing, the placement of the IQI for least favourable geometry.
    a. film side of the object
    b. source side of the object
    c. between the operator and the radiation side
    d. at the window of the X-ray tube.
    e. between intensifying screen and the films.

23. Typically used radiographic sources in the weld inspection.

    a. Cobalt 60
    b. Iridium 192
    c. X-ray
    d. All of the above
    e. B & C of the above

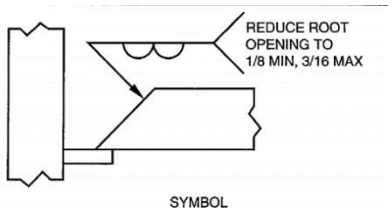
24. Material Ductility are related with?
    a. Hardness
    b. Toughness
    c. Tensile Test
    d. Conductivity

25. Flux is an essential variable for which welding process?

    a. FCAW
    b. SAW
    c. ESW
    d. GMAW

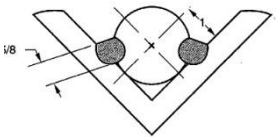
26. What following contibute under bead cracks ?
    a. preheat
    b. H2

27. What is indicate by this symbol?



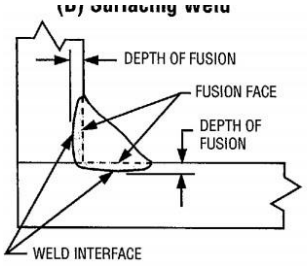
a. Surface welding
b. Reduce root opening
c. Weld bevel angle
d. Base Metal

28. which one is correct as per this figure?

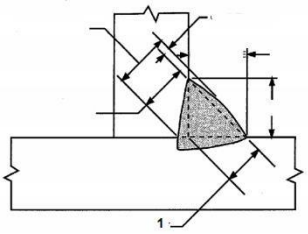


a. Spot weld
b. Pug weld
c. Flare weld
d. Slot weld

29. Which one is depth of fusion?



30. What is indicate by Number#1 in this symbol?



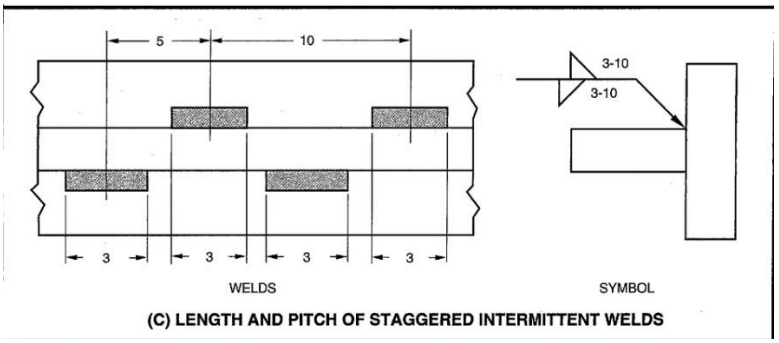
a. Weld Leg
b. Fusion Face
c. Weld Throat
d. Weld Interface

31. The technique where each individual weld passes is deposited in the direction opposite that of the overall progression of welding Is called?

a. Block sequence
b. Cascade sequence
c. Back step sequence

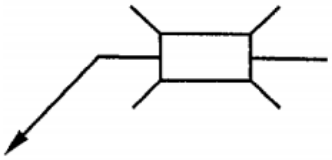
32. As shown figure what is the distance between weld.

a. 5
b. 3
c. 7
d. 10



33. This Figure is indicated?

a. Double Bevel groove with plug weld
b. Single V Groove with Plug Weld
c. Double V Groove with Spacer
c. Single Bevel Groove with Spacer



34. Which of following best describes electrode classification for CS Electrode for SMAW?
a) Electrode, atomic symbols of the major elements in the electrode in descending order of concentration
b) Electrode, hydrogen content , yield strength of resulting weld metal, and suitable welding position
c) Electrode, chemical composition of undiluted weld metal, suitable welding position, and type of suitable current
d) Electrode, minimum tensile strength, suitable welding position, type of covering and current

35. Which of the welding techniques describes a welding process in which the filler metal feeding is machine operated while the joint travel speed and guidance are the responsibility of the welder?
a) Manual
b) Semi Automatic
c) Mechanized
d) Automatic

36. In SMAW, an increase in arc length results in?
a) decreased current ; decreased voltage
b) decreased current ; increased voltage
c) increased heat input ; decreased voltage
d) increased current ; increased voltage

37. In Plasma Arc Welding, which discontinuity most related below?
a) Slag Inclusion
b) Copper inclusion
c) Overlap
d) Undercut

38. In Electro Slag Welding Copper Shoes are used for?
a) To avoid Slag inclusion
b) To prevent excessive heat of Base metal and block Weld metal
c) To reduce Electrode consumption
d) TO prevent Crack propagation

39. In CAC-A process, Metal removed by?
a) by Cleaning after cut
b) By Air study using compressor
c) by Wire brush
d) by solvent cleaning

40. MT can efficiently be done in?
a) Stainless weld on Stainless base metal
b) Carbon steel weld on Stainless base metal
c) Carbon steel weld on Carbon steel base weld
d) None of these

41. A welder continuously welds for 10 Minutes with 60 duty cycle to a complete A weld, when the welder welds using 100 percent duty cycle, how much time (m/s) is required to complete the weld?
a) 12 min
b) 60 min
c) 6 min
d) 1 hour

42. Transverse Bend Specimen shall be cut from?
a) Side Bend
b) Face Bend
c) Root Bend
d) Side, Face, Root bend

43. In Nick Break Testing, the specimen shall be broken by?
a) One end Fixed and other end hit by hammer
b) Put in Tensile Testing Machine
c) By Put in Bend Test Rig
d) None of the above

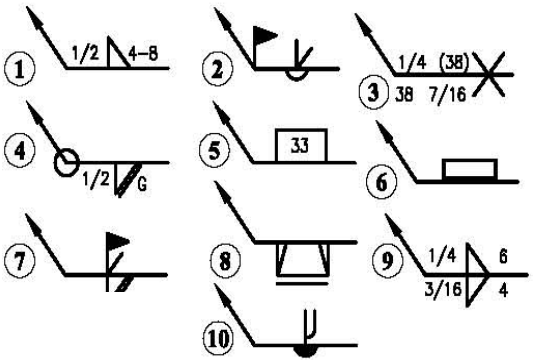
44. In welding sequence, Welds done in both side and the multiple layers is done as Overlapping?

a) Block Sequence
b) Cascade Sequence
c) Split layer
d) Back step sequence

45. Stud welds+ Surfacing welds always should be done in?
a) Both side
b) Opposite to weld side
c) Arrow side
d) groove side

Note: Surfacing and Stud weld symbol does not indicate the welding of a joint, so it has no Arrow side or other side significance.

46. In Intermittent welds, dimension in the order of?
a) Length, pitch, Size
b) Pitch, Length, Size
c) Size, Pitch, Length
d) Size, Length, Pitch



47. In Figure Symbol 10 represents?
a) Stud weld with Melt through
b) Fillet Weld with Melt through
c) Plug weld with Melt through
d) Groove weld with Melt through

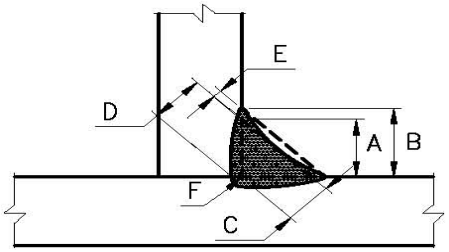
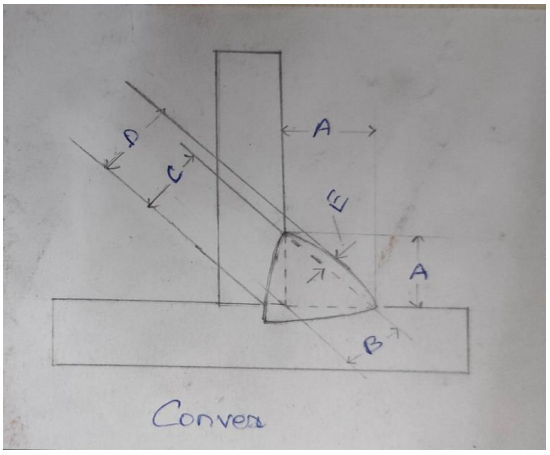


Fig. 16

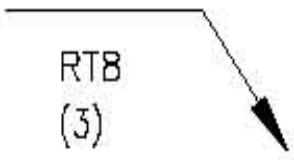
48. In Figure 16, the dimension C represents?
a) Actual throat
b) Weld size
c) Concavity
d) Theoretical throat

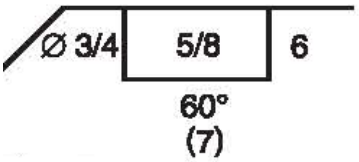


49. In Figure 16, the dimension E represents?
a) Concavity
b) Actual throat
c) Convexity
d) Size

50. In Figure, the dimension A represents?
a) Convexity
b) Effective throat
c) Theoretical throat
d) Size and Leg

51. The welding symbol shown below represents?
a) 8 RT every 3”
b) 3” long RT for every 8”
c) 3RT for with pitch of 8”
d) RT for 8” at 3 random locations





52. In Figure the dimension 5/8 represents (inside the weld symbol)?
a) 5/8 stands for groove length
b) 5/8 stands for size of slot
c) 5/8 stands for depth of Filling
d) 5/8 stands for groove width

53. When two pipes welded together invertical axis which represents the Position of?

a) 1G
b) 2G
c) 5G
d) 6G

54. The Absorption of energy, which is related to?
a) Toughness
b) Fatigue
c) Hardness
d) Tensile

55. Which NDE test is used to determine the soundness of the weld except NICK Bend test?
a) UT, RT
b) PT, MT
c) PT, RT
d) MT, UT

56. If Preheat has not done on the weld, what will happen below?
a) wider Heat affected zone
b) avoid delayed cracking
c) improves weld quality
d) Narrow heat affected zone

57. Porosity is defined as?
a) Linear type of Defect
b) Transverse type of defect
c) defect perpendicular to stress
d) cavity type of Defect

58. Incomplete Fusion is defined as?
a) Fusion does not occur between base metal and weld metal
b) Flux entrapped in the weld metal
c) Cavity type o defect
d) weld metal is poured onto base metal

59. Inadequate Penetration occurs due to?
a) High heat input
b) Low thickness of base metal
c) Improper electrode

60. Under fill is known as?
a) Excess filling of weld groove
b) Convex shape of reinforcement
c) complete removal of reinforcement
d) inadequate filling of weld groove

60. Weld Reinforcement is defined as?
a) Mating surface is come to close contact
b) the Point where Base metal and Weld metal joined
c) Excess metal is used to fill the weld groove
d) The line separating weld metal and Base metal

61. Which defect will be formed on Arc strike?
a) Porosity
b) Undercut
c) Overlap
d) Crack

62. Which NDE method is used extensively?
a) RT
b) UT
c) MT
d) VT

63. In PT Testing, After Dwell what action will be done?
a) Apply the Penetrant
b) Apply the developer
c) Remove the excess penetrant
d) Cleaning the part

64. Capillary Action is used in which NDE Method?
a) MT
b) RT
c) PT
d) UT

65. Advantages of UT over RT?
a) Both side access needed
b) one side access required
c) calibration required
d) couplant required

66. In below which Isotope is used in RT?
a) Iridium Ir192
b) Carbon
c) Hydrogen
d) Iron

67. The substance used in UT to aid in the transmission of sound from the search unit to the work piece and back is called?
a) Solvent
b) Couplant
c) Cable
d) Transducer

68. In Electro Slag welding process utilizes what type of orientation with welding occurring in the Flat position?
a) Horizontal
b) Flat
c) Vertical
d) Overhead

69. Braze joints are welded together in which joint below?
a) Edge
b) Fillet
c) Butt or Lap joints
d) Corner

70. The Brazing is differing from other metal joining process? a) Base metal will not melt filler metal should melt below 840°F
b) Base metal melts & filler metal will not melt more than 840°F
c) Base metal melts filler metal melts more than 840°F
d) Base metal will not melt, filler metal should melt more than 840°F

71. The condition under which the CWI can publicity express an opinion on welding inspection subjects under the AWS QC1 code of ethics is?
a) When it is the consensus of the inspection department
b) When requested by a legitimate news organization
c) When the CWI strongly feels an opinion is justified
d) When founded upon knowledge of the facts in issue

72. In Radiography testing, the placement of the IQI for least favorable geometry?
a) Between intensifying screen and the films.
b) At the window of the X-ray tube.
c) source side of the object
d) Film side of the object

73. In which of following ways are fluxes for SAW classified according to manner in which they modify. The composition of the weld metal?
a) Alloyed (or) Neutral
b) Neutral (or) Agglomented
c) composited (or) Fused
d) Fused (or) agglomented

74. In which method granular flux can be applied during welding?
a) FCAW
b) GTAW
c) SAW
d) GMAW

75. The rapid quenching (increasing the cooling rate) of high carbon steel from austenitizing range will result in formation of:

a) Austenite
b) Pearlite
c) Cementite
d) Martensite

76. In a Single V-Groove weld, the term for the sloped surfaces against which the weld metal is applied is?

a) Root face
b) Joint root
c) Groove face
d) Root edge

77. In groove weld cross section the line separating weld metal from base metal is called?

a) Fusion face
b) Depth of fusion
c) Weld interface
d) Face reinforcement

78. Stainless steels are defined as having at least what percent chromium?

a) 7%
b) 15%
c) 30%
d) 12%

79. An E71T-1 electrode designation is for which welding process? a) PAW

b) SMAW
c) GMAW
d) FCAW

80. FCAW electrode classification uses the letter “T”?

a) Tungsten
b) Titanium
c) Tubular
d) Thorium

81. Limitation of the oxy fuel cutting process?
a) it takes longer to cut thin steels than with other cutting processes
b) process is limited to readily oxidized material
c) large plate sections take several passes to cut
d) it is hard to change direction using OFC

82. Usually in which of the form the filler metals for manual GTAW are used?
a) 12 inch (0.3 meters) coated electrodes
b) Coils of tubular electrode
c) Large wire spools (30 lbs)
d) Cut lengths (rods) usually 0.91 meters (36 inches) long

83. In GMAW Welding Process, CO_2 is used as shielding gas, because?
a) it has high mixing ratio
b) it has stability in high pressure
c) it has light density
d) it has Low cost,Penreration

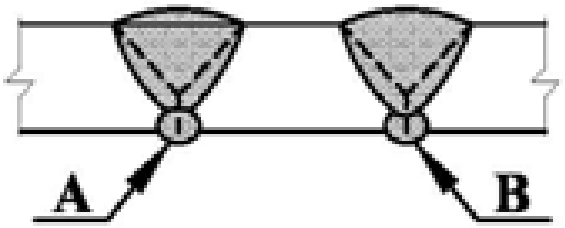
84. The distance between the tip of the welding electrode and the weld pool surface is called?
a) arc strike
b) arc blow
c) arc length
d) arc force

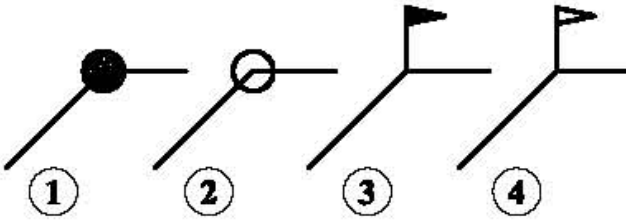
85. If a welding inspector feels he or she is not qualified to make judgment as to the acceptability of weldment, he or she should?
a) Allow work to continue because further welder experience will improve quality
b) wait to make judgment until he or she had time to get additional training
c) Do the best he or she can and report accordingly
d) Call for assistance from someone who is qualified

86. NDE technicians are usually qualified in accordance with the requirement of?
a) ASME SECTION IX
b) AWAS D1.1
c) ASME SECTION VIII
d) ASNT SNT-TC-1A

87. Which of the following results from improper termination of the SMAW electrode and shrinking of the molten weld pool during welding?
a) incomplete fusion
b) porosity
c) slag inclusion
d) crater crack

88. The rapid quenching of high carbon steel from austenitizing range will result in formation of?
a) Austenite
b) Pearlite
c) Bainite
d) Martensite


A
B


1
2
3
4

89. Ability to be cyclically loaded without failing is related to which of the following properties of A Metal?

a) Hardness
b) Tensile
c) Fatigue
d) Ductility

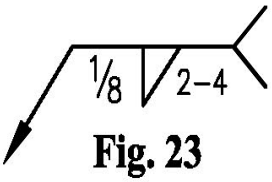


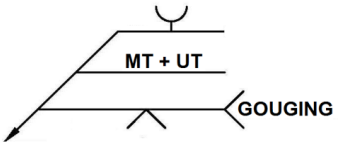
Fig. 23

90. The figure 23 shows what welding symbol?

a) Intermittent 1/8-inch fillet welds 4 inches long on 2-inch centers
b) Intermittent 1/8-inch fillet welds 2 inches long on 4-inch centers
c) Two 1/8-inch fillet welds on 4 inches centers
d) Four 2-inch fillet welds 4 inches long at 1/8 inch spacing

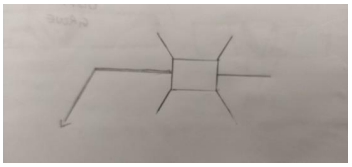
91. In Figure, what action should be done first during welding?

a) welding on other side
b) MT test on Arrow side
c) Gouging the root surface
d) WELDING V groove



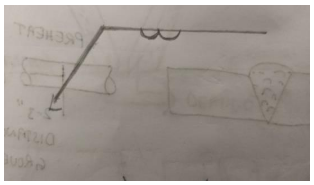
92. In Figure, the welding symbol shows?

a) Backing bar
b) Backing weld
c) Consumable insert
d) Spacer

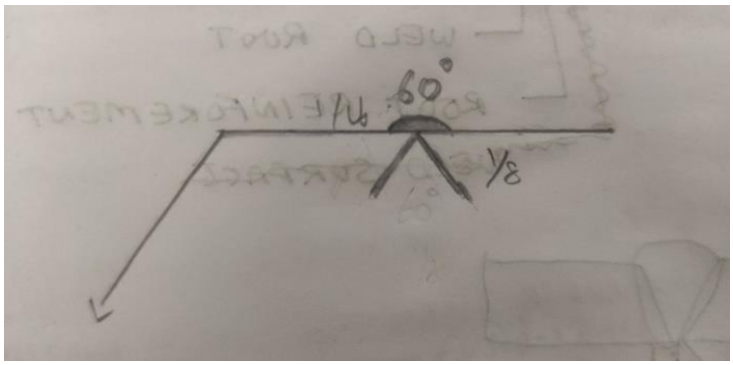


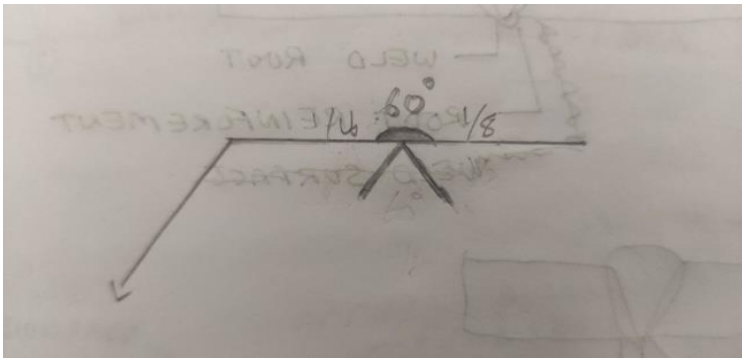
93. In Figure, the welding symbol shows the type of weld is?

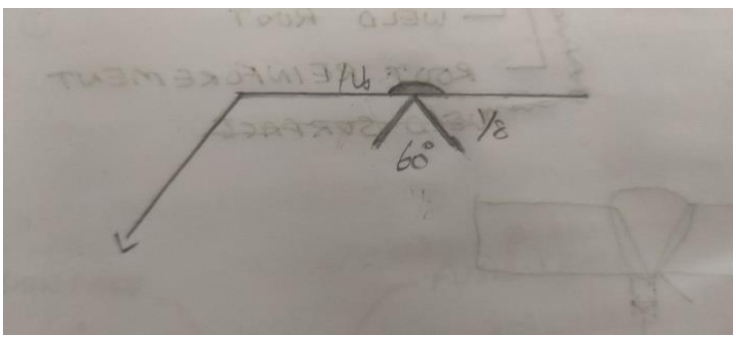
a) Edge Weld
b) Back Weld
c) Backing weld
d) Surface weld

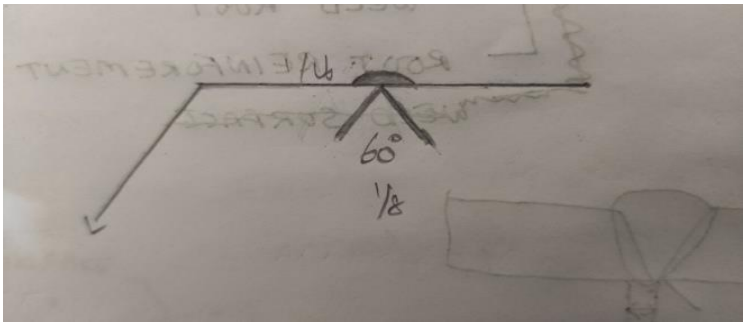


94. In Figure, which welding symbol shows that V groove weld is having 60° groove angle, 1/8 inch root opening and 1/16 inch size Melthrough?









Q AND A PART A CALCULATION

1. Bevel angle is 37.5 with tolerance of 2.5 +- what is the acceptance range ?
   35 to 40
2. Find 25% for 20
   5
3. A Welder continuously held for 10 mts with 60 duty cycle to complete a weld by using 100% duty cycle how much minutes it will take ?
   6 Minute
4. 40 circumferential tolerance 20% what is the total tolerance rate
   8
5. Post weld heat treatment temperature cools down 1200 to 1100 degree in 30 mts what is the cooling rate in degree per hour
   200
6. In GMAW welding process the 12 inch length as welded in 10 secs in 1 minute how much length can be welded
   72
7. Root opening is 1/8 th inch allowable tolerance + 1/16 -1/32whats the range ?
   3/32 to 3/16
8. For 20 inch dia pipe 40 percent is welded to remove the clamp . How much length welded ?
   25
9. Qualification range .5 to 1.1 t value of T is .76 what is the qualification range
   .38 to .836
10. For 20 mm given range is given tolerance +-10% what's the range ?
    18 to 22
11. Heat input formula = $\frac{\text{voltage X AmpereX 60}}{\text{Travel speed}}$

What heat treatment is characterized by holding the part at the austenitizing temperature for some time and then slow cooling in the furnace?

Normalizing
Quenching
Annealing
Tempering
Stress relief

What heat treatment characterized by holding the part at austenitizing temperature and then cooling in still air?

a) Normalizing
b) Quenching
c) Annealing
d) Tempering
e) Stress relief

Who is normally responsible for the qualification of welding procedures and welders?

Welder
Architect
Welder's employer
Engineer
Code body

What weld metal discontinuity results when the welder fails to completely fill the weld groove?

a) Underfill
b) Undercut
c) Overlap
d) Incomplete fusion
e) None of the above

Which of the following methods for controlling materials is a fabrication shop, is most suitable for automation?

Color coding
Alpha-numeric coding
Heat number transfer
Bar coding
Segregation by alloy

What weld discontinuity results when the welder travels too slowly, causing excess weld metal to flow out of the joint and lay on the base metal surface without fusing?

a) Undercut
b) Under fill
c) Overlap
d) Incomplete fusion
e) None of the above

The discontinuity that results from the entrapment of gas within the weld cross section is referred to as:

a) Crack
b) Slag inclusion
c) Incomplete fusion
d) Porosity

A discontinuity is:

a) Always a defect
b) Always a reject
c) Always acceptable
d) Rejectable if it exceeds code limits
e) None of the above

That portion of single bevel butt joint where the pieces to be joined come closet together is referred to as the?

Radius
Joint root
Bevel angle
Grove angle

The type of joint formed when the two pieces to be joined lie in parallel planes and their edges overlap is called?

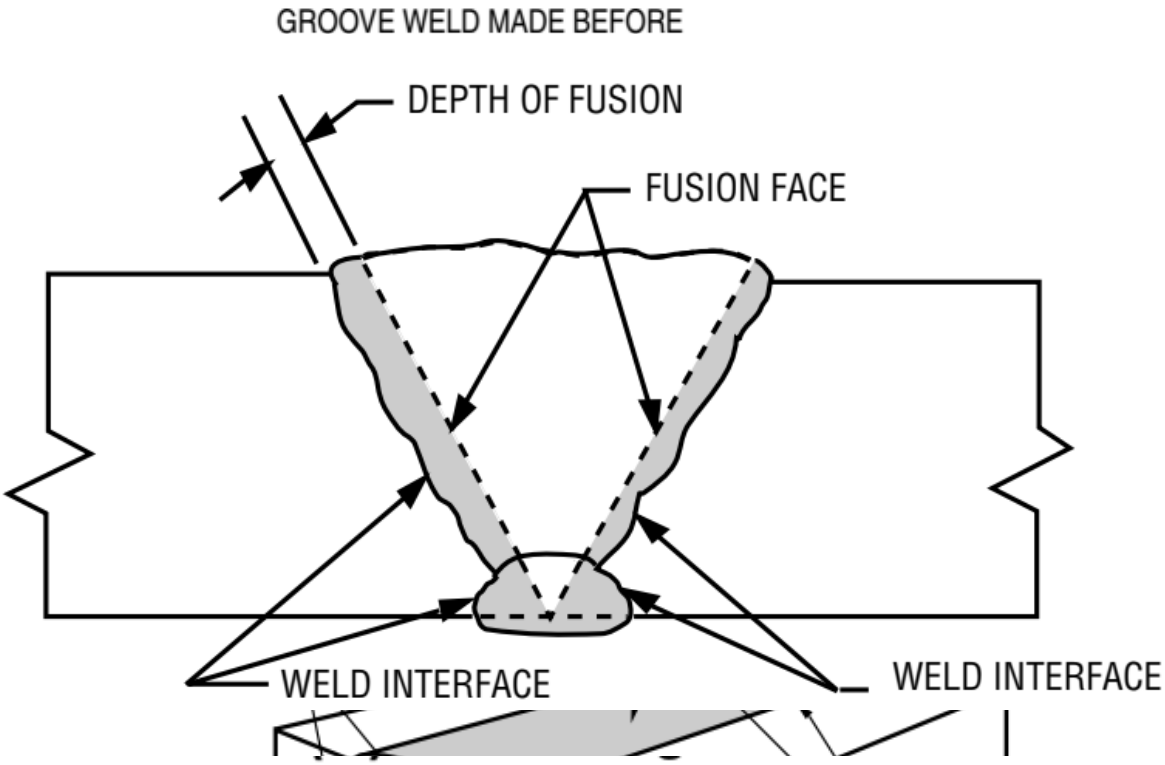
Corner
T
Edge
Lap
Butt

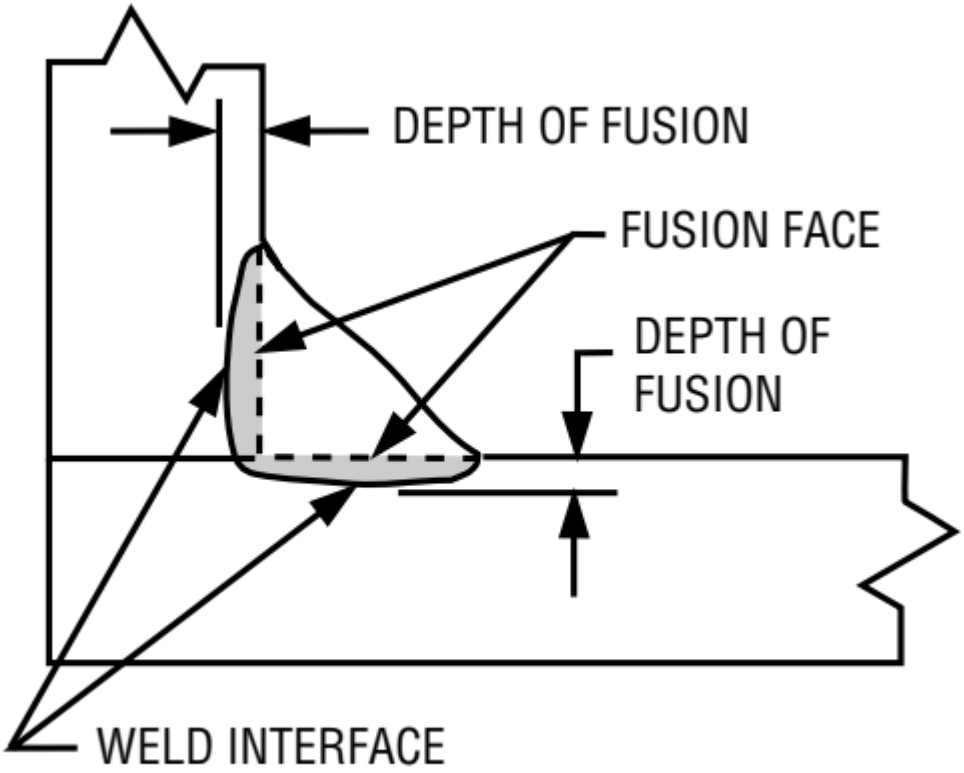
Which of the following are not common to both GTAW and PAW?

Non-consumable tungsten electrode
Constricting orifice
Shielding gas nozzle
Externally applied filler metal
None of the above

NDE personnel (other than visual) should be certified to what document?

QC – 1
D1.1
API 1104
ASNT – TC – 1A





(C) Fillet Weld

Weld Terms

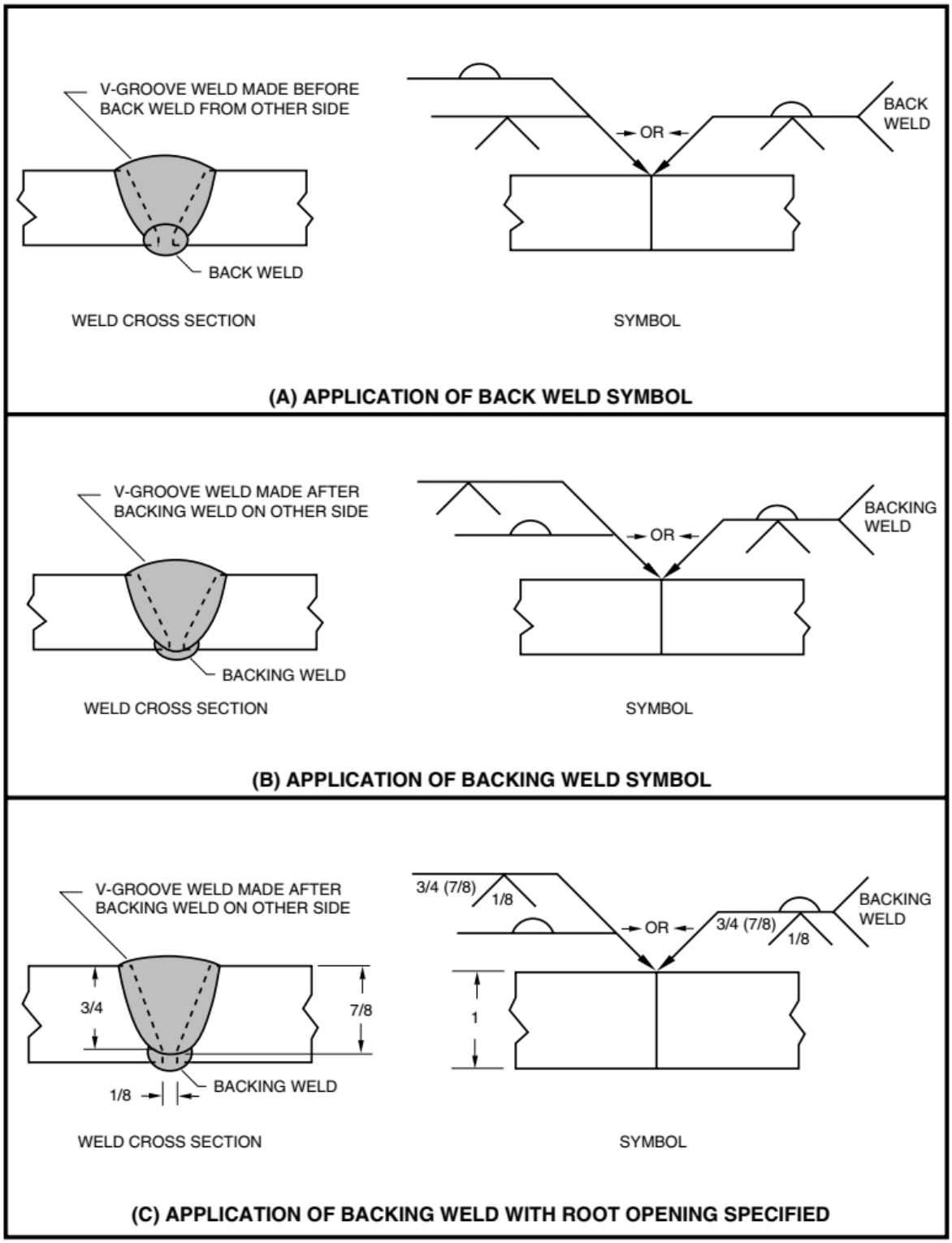
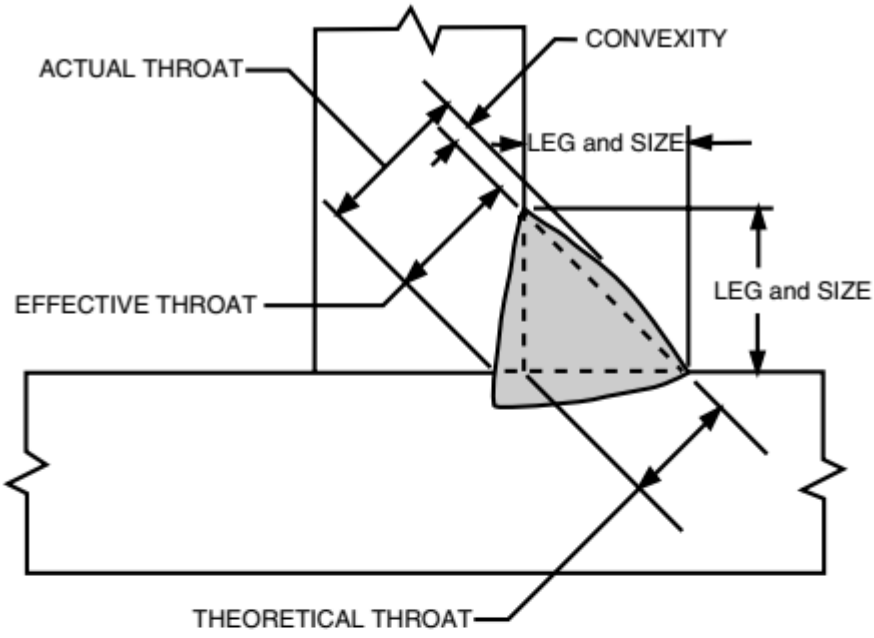
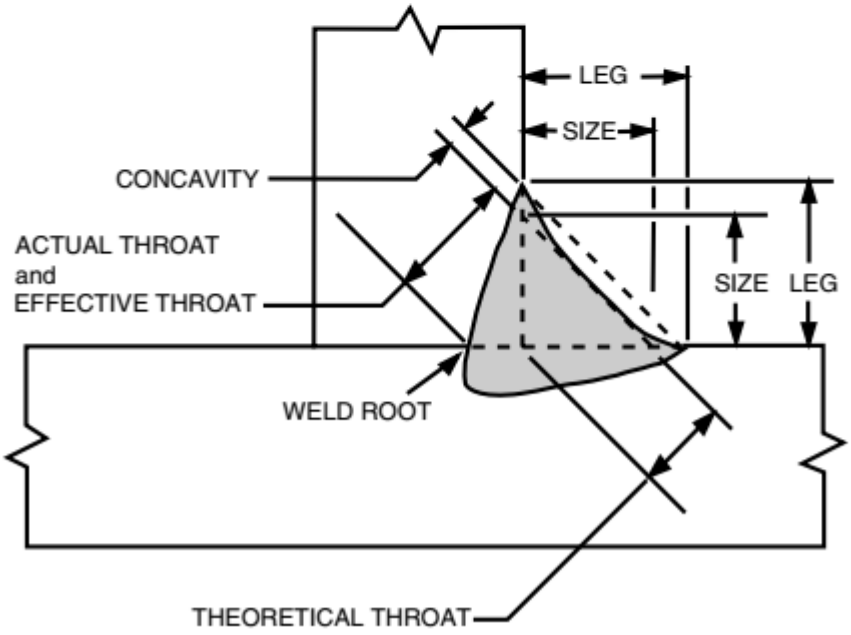


Figure 4.98 —Applications of Back or Backing Weld Symbol



CONVEX FILLET WELD

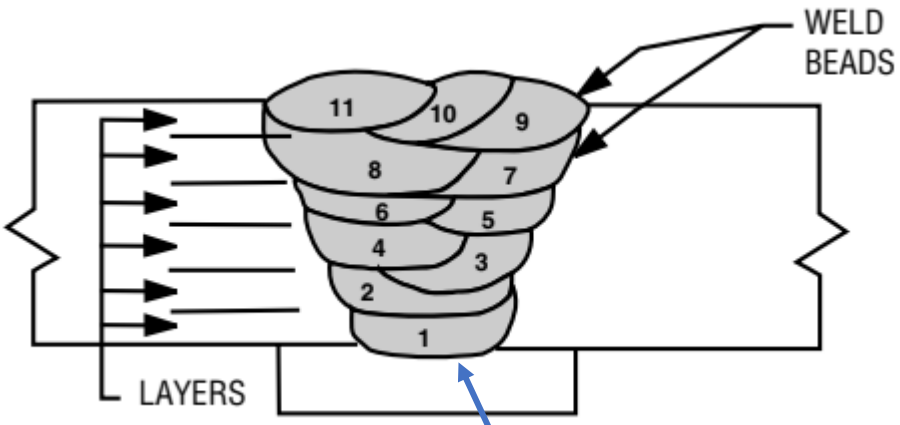


CONCAVE FILLET WELD

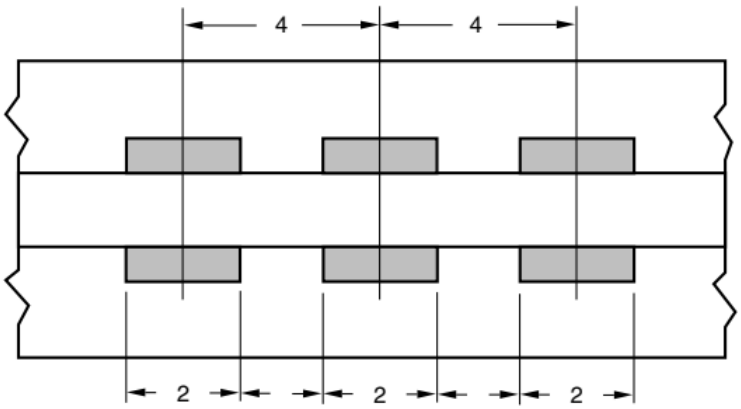


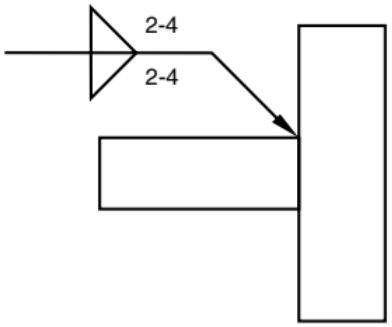
(C) SINGLE-V-GROOVE WELD

Melt-Through



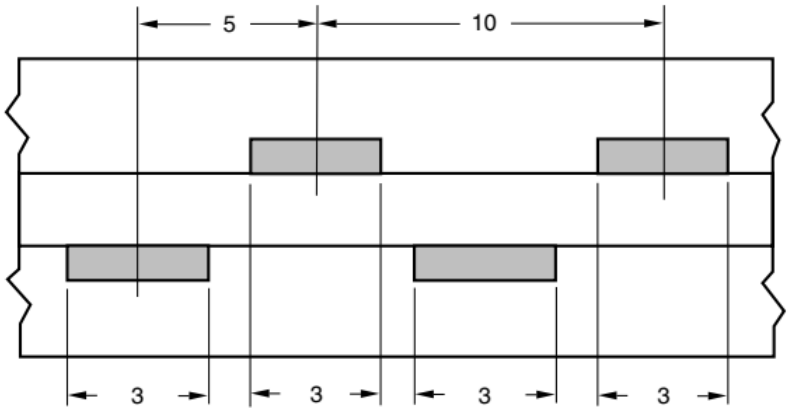
ROOT SURFACE

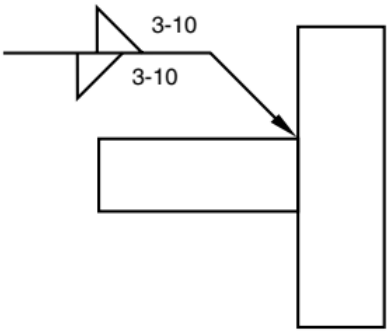




WELDS SYMBOL

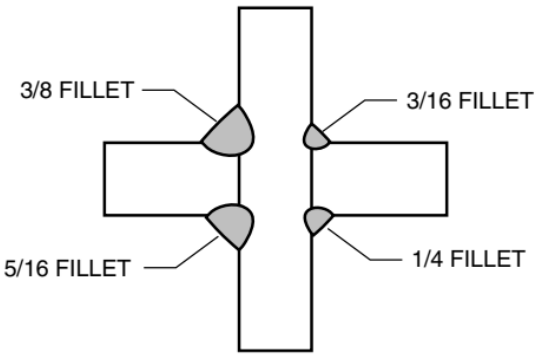
(B) LENGTH AND PITCH OF CHAIN INTERMITTENT WELDS

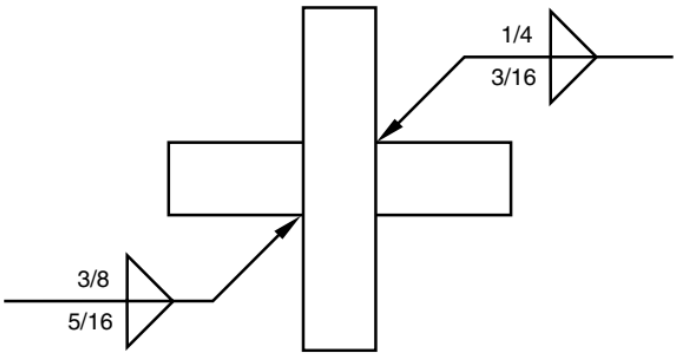




WELDS SYMBOL

(C) LENGTH AND PITCH OF STAGGERED INTERMITTENT WELDS





(C) DOUBLE-FILLET WELD SYMBOLS FOR TWO JOINTS

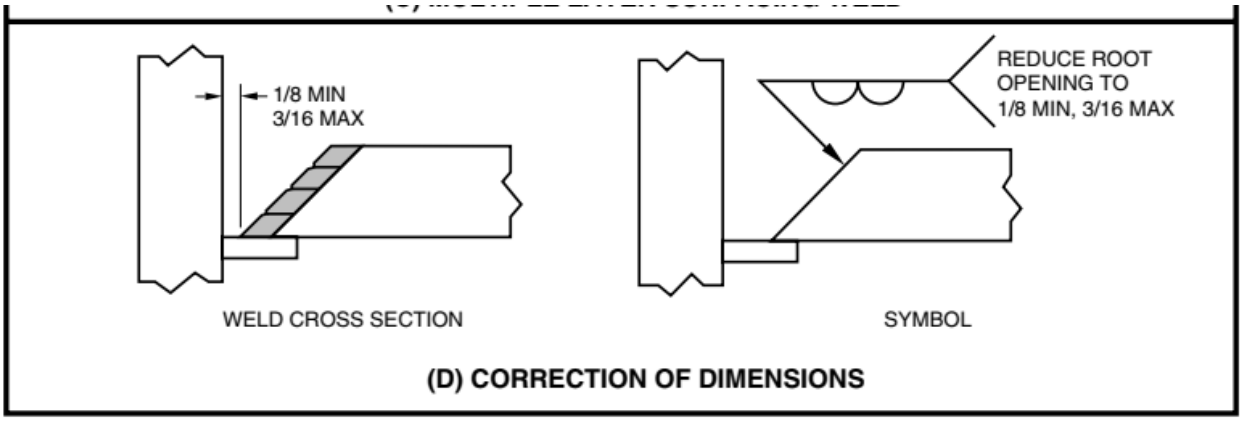


Figure 4.80—Surfacing Weld Dimensioning

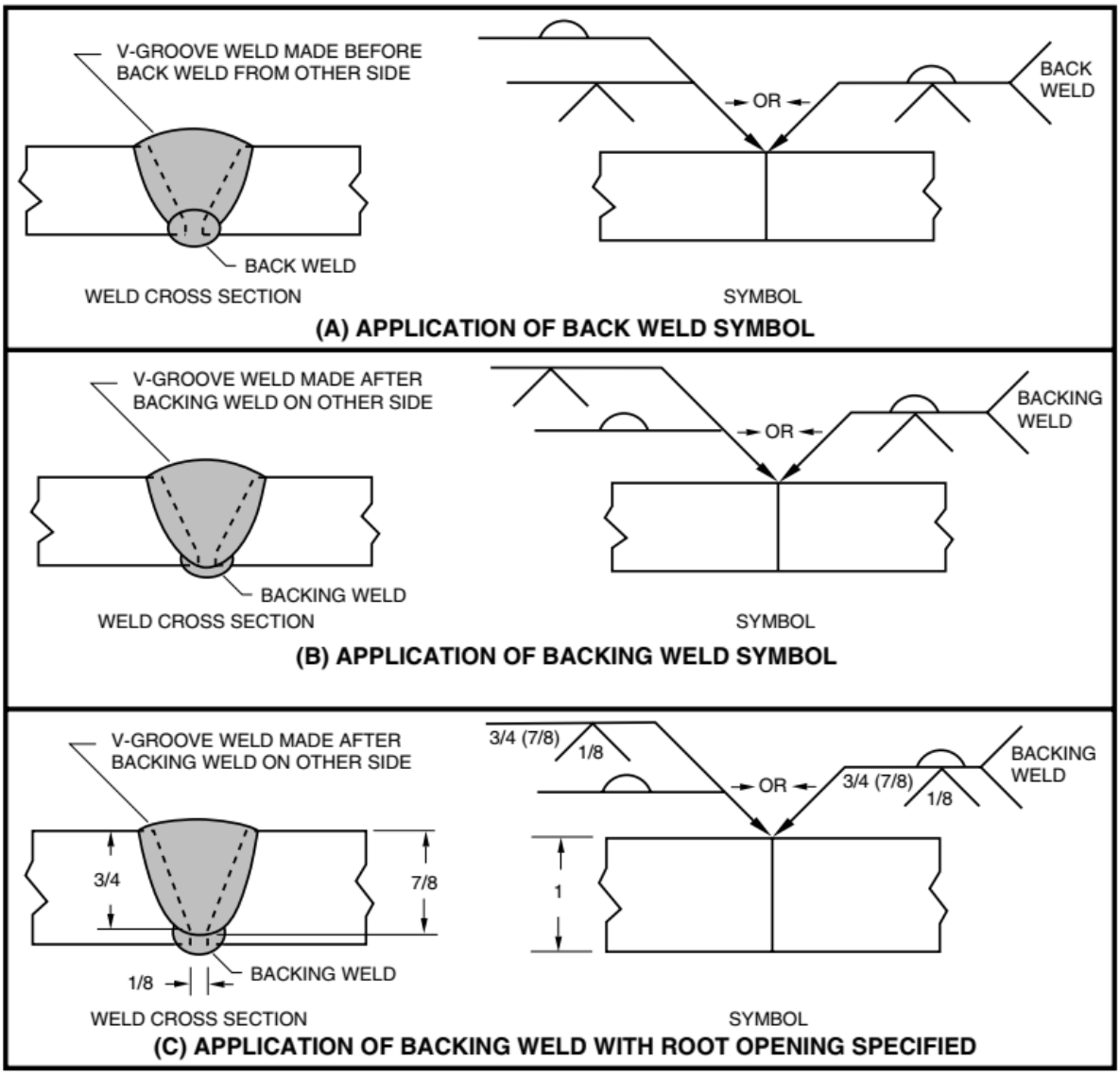
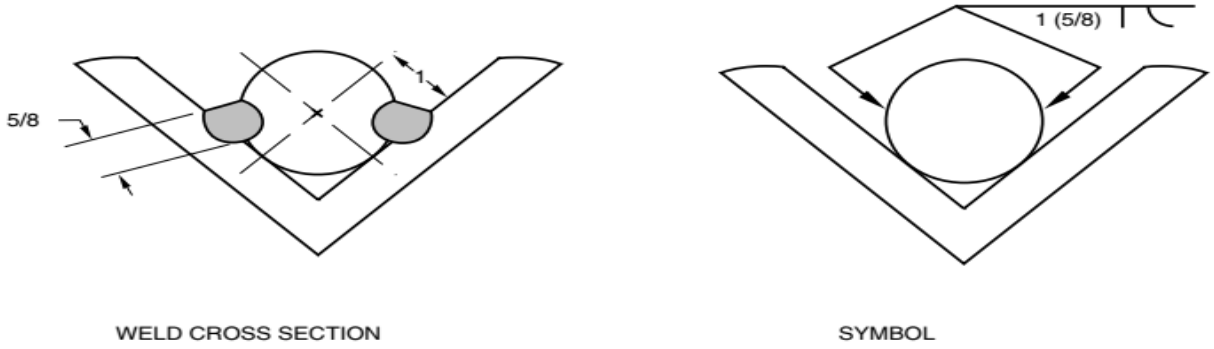
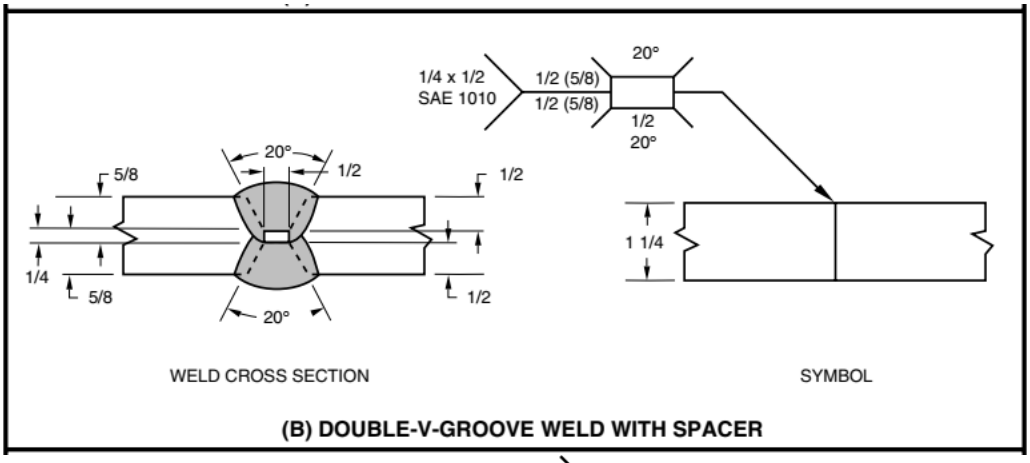


Figure 4.81—Application of Back and Backing Weld Symbols



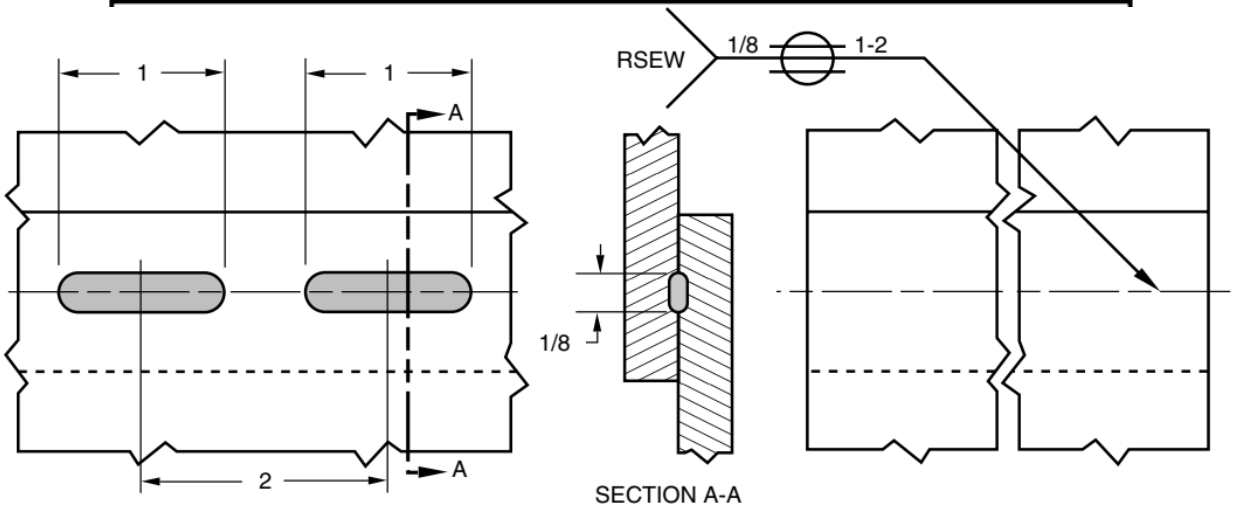
WELD CROSS SECTION SYMBOL

(F) TWO SINGLE-FLARE-BEVEL-GROOVE WELDS



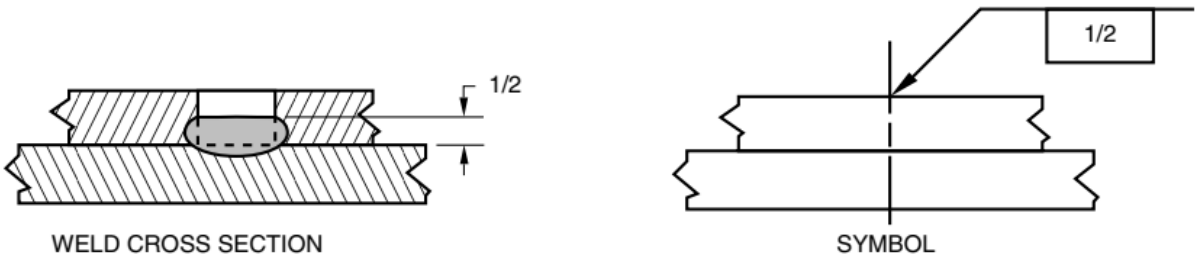
WELD CROSS SECTION SYMBOL

(B) DOUBLE-V-GROOVE WELD WITH SPACER



WELD CROSS SECTION SYMBOL

(A) SIZE, LENGTH AND PITCH OF INTERMITTENT SEAM WELDS



WELD CROSS SECTION SYMBOL

(C) DEPTH OF FILLING

OVERLAP

" The protrusion of weld metal beyond the weld toe or weld root "

backing weld—Backing in the form of a weld, made before the primary weld [see Figure 4.19(F)].

back weld—A weld made at the back of a single groove welded joint after the completion of the groove weld [see Figure 4.19(E)].

backstep sequence—A longitudinal sequence in which weld passes are made in the direction opposite to the progression of welding [see Figure 4.35(A)].

face reinforcement—Weld reinforcement on the side of the joint from which welding was done [see Figure

joint root—That portion of a joint where the members approach closest to each other. When viewed in cross section the joint root may be either a point, a line, or an area (see Figure 4.12).

weld toe—The junction of the weld face and the base metal [see Figure 4.21(B)].

root opening—a separation at the joint root between the workpieces [see Figure 4.14(A) and (E)].

خلي بالك هتلاقي ف الاختيارات ROOT GAP لا تختارها لانها غير ستاندر

root surface—The exposed surface of a weld opposite the side from which welding was done [see Figure 4.21(C)].

faying surface—The mating surface of a member that is in contact with or in close proximity to another member to which it is to be joined.

edge joint—A joint between the edges of two or more parallel or nearly parallel members [see Figures 4.2(E), 4.3(E), and 4.11].

جايب التعريف ده بس مجبش في الاختيارات الـ EDGE JOINT ولكن جاب مكانها الـ PARAELLE JOINTودا المسمى الموجود في مواصفة التعريفات A3.0 في اخر اصدار.... انت اختار PARAELLE JOINT

hold point—a specific, prearranged step in the fabrication process where fabrication is stopped to permit an interim inspection. Fabrication can begin again only when the inspection shows the part meets the quality requirements.

piping porosity. A form of porosity having a length greater than its width that lies approximately perpendicular to the weld face. See also **elongated porosity**.

خلي بالك هجيب في الاختيارات linear prosity و wormhole كل دا غلط انت هيجي التعريف ده تختار PIPING POROSITY

wormhole porosity. A nonstandard term when used for piping porosity

collar. The reinforcing metal of a nonpressure thermite weld.

slugging. The unauthorized addition of metal, such as a length of rod, to a joint before welding or between passes, often resulting in a weld with incomplete fusion.

- RT film sensitivity

- بيقول ايه اللي بيتحكم في sensitivity Film اخترت الـ grain Size لان كلما صغر حجم الحبيبات كلما زادت دزدة التصوير(D4,D7)
- قال في لحام الSAW لو المشغل بيقدر يغير في متغيرات اللحام اثناء اللحام تبقي ايه AUTOMATC-MECHANAZIED -ROBOTIC
- ايه اللي بيقيس قطر البروستي الدائرية انا اخترت البرجل
- جه سؤال بيقول لو عملت اختبار MT بين اتنين ماتريال مختلفين في المغناطيسية انا اخترت Non relevant indication
- سأل عن ان WPS فيه VARAIABLES ليه بنهتم بيها ومن ضمن الاختيارات اتنين بس اللي يلخبطوا واحد بيقول لضمان جودة الجزء اللي بيتم تصنيعه والاختيار التاني لضمان جودة والخواص الميكانيكة للحام انا اخترت الاختيار التاني لان WPS خاص بوصلة اللحام وليس كل الComponent اللي بيتم تصنيعه
- سأل عن ماذا يحدث لو معملتش Preheat بالشكل المطلوب للحام STUD WELD من ضمن الاختيارات انها هتقلل عرض منطقة ال HAZ.. ده عموما لاي PRHEAT معرفش الـ STUD WELDهيكون له حاجة مميزة في الموضوع ده ولا لا

- وجاب رسمة عليها ادوات وقال ايه الاختبار اللي بيتعمل علي STUD WELD دي الجزئية اللي بتتكلم عنها من الكتاب

some other process. Once welded, a stud can be easily inspected. First a visual examination is made to assure the presence of a 360° flash. Then the stud can be either struck with a hammer or pulled to judge its acceptability. When struck with a hammer, a good stud weld will ring while a poor joint will result in a dull "thud." If the stud is threaded it can be torque tested to determine its quality.

-IONIZED OR ORFICE في الـ PAW

melt-through. Visible root reinforcement in a joint welded from one side. See also root reinforcement and root surface.

- رسمة GMAW (SPRAY)
- ضغط اسطواتة الاسيتلين 15 PSI
- معنى 308فى سلكة 308L
  معناها ALLOY
- ما الذى يميز GMAWعن SMAW.... DEPOSTION RATE
- التهوية للحام تكون LATERALY ACROSS
- اى من الانواع التالية ليست SOLDERING
  الاجابة COLD SOLDERING
- GMAW بديلة للـ SMAW
- 7016 6 تعنى LOW HYDROGEN
- ER R تعنى ROD
- Tيعنى TUBLER
- يتم اختيار الـ ELECTROD فى GTAW طبقا للـ COLOR CODE
- الاعتبارات عند اختيارات ال FILLER METAL ان يكون تكوينها الكميائى مثل الـ BASE METAL
- بتقيس البروستي بالبرجل
- ماذا يتج عن الفشل في ملىء اللحام والاختيارات فيها UNDERFILL و CRACK انا اخترت UNDERFILLبالرغم انها تمشي كراك

بتعمل تأهيل للحام ونجح في كل تاهيالت الكود ايه الحاجه إلضافية التي يفضل تكون موجودة عند اللحام ؟
efficiency – **Workmanship** – deposition rate
-ماهى ميزة الShielded Self FCAW
minimize flames – not need Shielding Gas
-جاب سؤال Fcaw g وقال معناها اي
-لو في inclusion Copper يبقى process : إلجابةPAW :
-سؤال عن ال PAC) CUTTING ARC PLASMA) : ايه من المواد اللى يتم قطعها افل من OFC إلجابة AL(Aluminum)
-مع ال Cutting Arc Carbon محتاج ايه إلزالة المصهور ؟ إلجابةAir Compressed
-motion weld metal and slag path-وظيفتها ماهى- Coper shoes
-Accuracy : both eyes with or without-correction -
-لو عندك joint Brazing ايه ال joint common most المستخدمة ؟
lap – T – Corner - Butt
-test Spark -بيستخدم اليه
malleability/chemical analysis/ductility

-لكن LT السؤال بيقول افضل حاجة تستخدمها عشان ال molecular اصغر حاجه وحاطط water
deionized argon air helium

-
سؤال عن سبب ال undercut وحاطت في االختياراتspeed travel : **amperage High**
-
Reduction of slag inclusion
increase heat input-
decrease voltage
decrease current

-Welder qualified smaw can weld SAW
NO.

- ceramic ferrule هو Stud Weld في بيستخدم إللي non consumable
-When weld is stronger than base metal during bend test will bend uniformly

-سال على عمل الshoes copper في الESW
joint design requirement / cooling rate

- رسمة ل Probe UT طالع منها Waves كتيرة بزوايا مختلفة وبيقولك ده ايه في االختيارات PAUT